

CLOUDS OF PROBABILITIES

Discoveries in the 1920s revealed that atomic structure is more complex than the simple Bohr model. The modern model shows that electrons occupy a series of “orbitals”—shells and subshells with a variety of shapes. As it is impossible to know all of their properties at a single instant (see pp.42–43), it is more accurate to think of these orbitals as fuzzy regions where the electrons are likely to be found—for some purposes, the electron’s properties are effectively “smeared out” across the orbital.

ORBITAL ARRANGEMENT

The three p-orbitals are arranged at right-angles to each other.

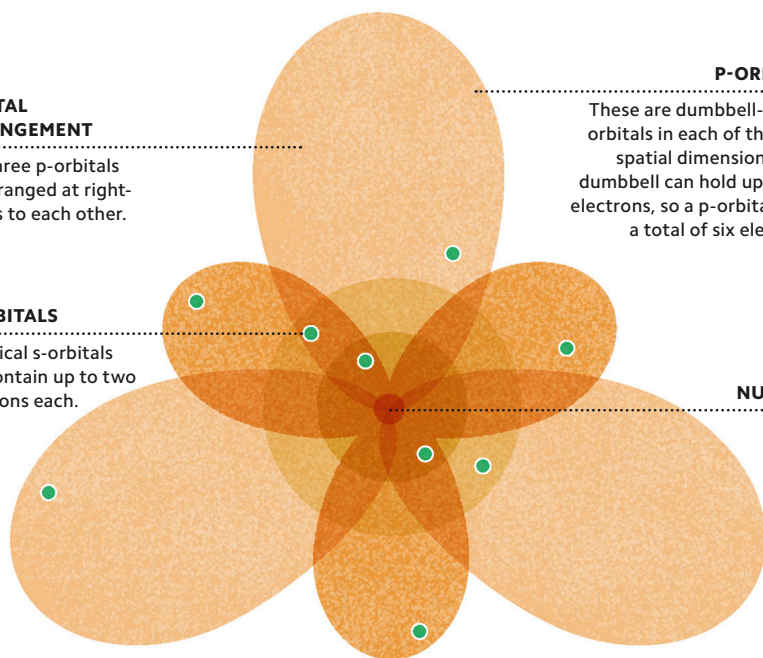
S-ORBITALS

Spherical s-orbitals can contain up to two electrons each.

P-ORBITALS

These are dumbbell-shaped orbitals in each of the three spatial dimensions. Each dumbbell can hold up to two electrons, so a p-orbital holds a total of six electrons.

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Orbiting a fluorine atom

A fluorine atom contains nine electrons, two each in its inner two S-orbitals and five in the first p-orbital.